LECTURE 2

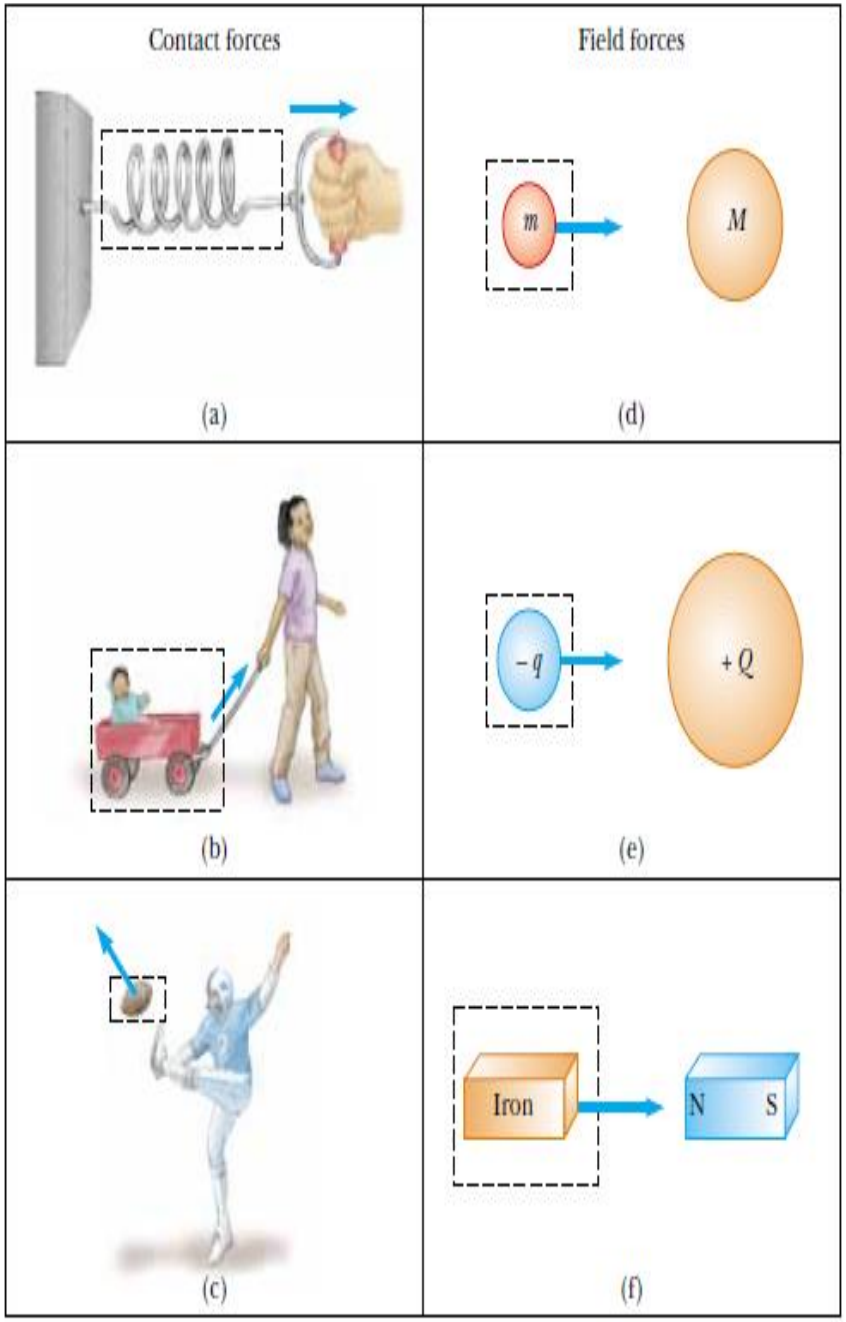
Newton Laws of motion



* The Concept of Force:

- ❖ It is an interaction with an object.
- ❖ It doesn't always cause motion.
- ❖ It is a vector quantity.
- ❖ It has SI unit of Newton (N) where $1N = 1\ Kg.m/s^2$

- It has many kinds.
- If there are several forces act simultaneously on an object Then,
- The **net force** acting on an object is defined as the vector sum of all forces acting on the object.
- If the net force exerted on an object is **zero**, the acceleration of the object is **zero** and its velocity remains constant.



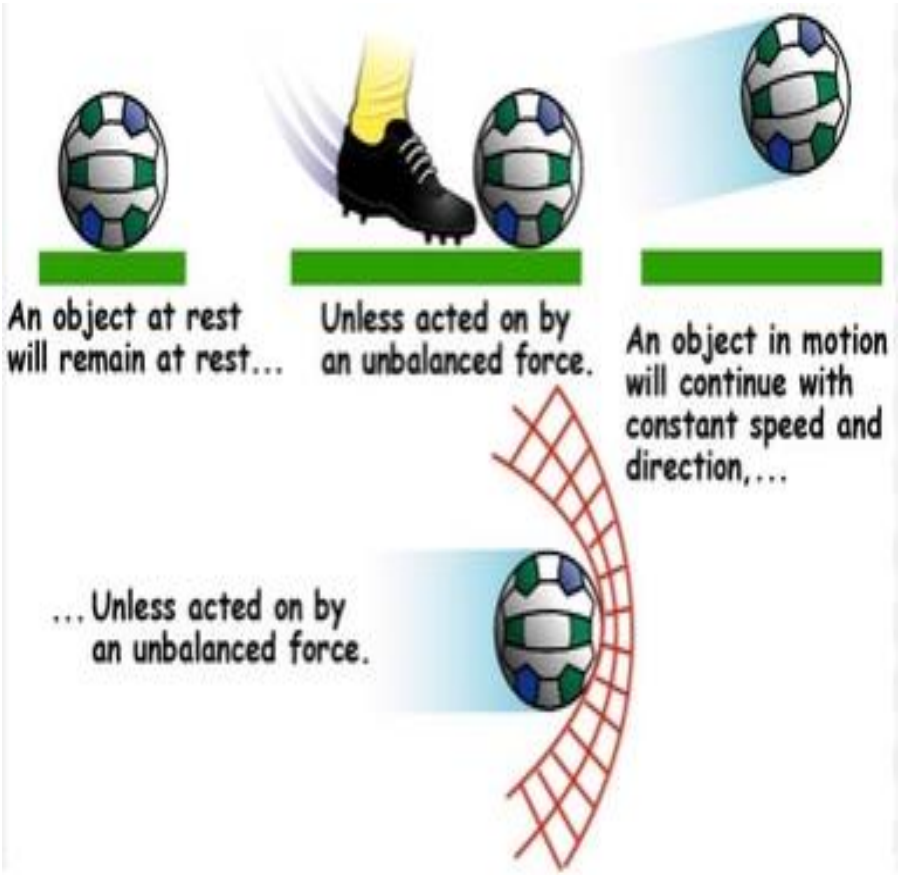
Newton's First Law:

❖ It also called " **The law of inertia**"

Inertia : is the tendency of an object to resist any attempt to change its velocity.

❖ It states that :

"In the absence of external forces, an object at rest remains at rest and an object in motion continues in motion with a constant velocity (that is, with a constant speed in a straight line"



Mass



❖ The bowling ball is more resistant to changes in its velocity than the basketball

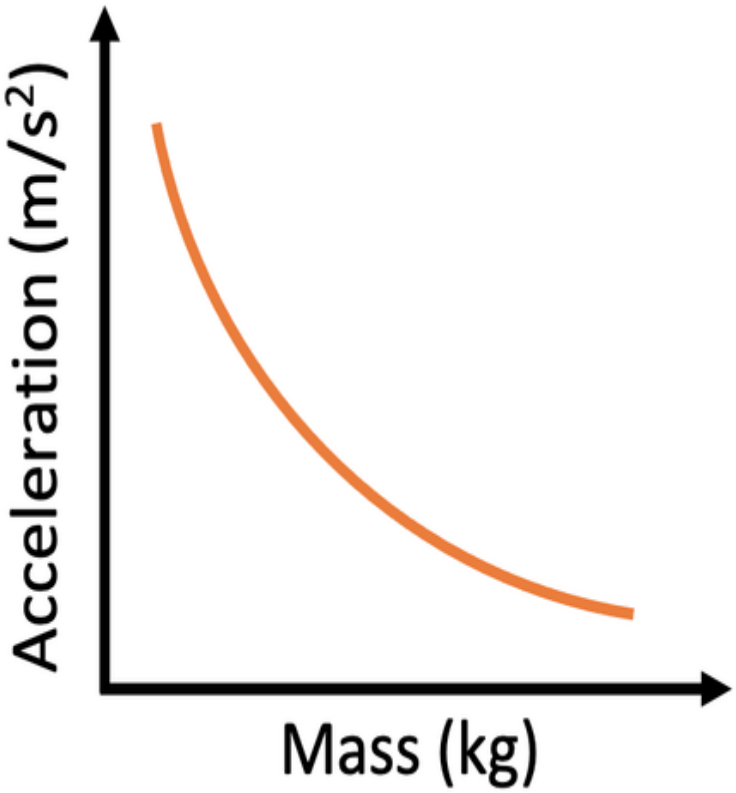
❖ Mass is that property of an object that specifies how much resistance an object exhibits to changes in its velocity.

❖ the SI unit of mass is the kilogram (Kg).

❖ mass is a scalar quantity

❖ The greater the mass of an object, the less that object accelerates under the action of a given applied force.

$$\frac{m_1}{m_2} \equiv \frac{a_2}{a_1}$$



The Gravitational Force and Weight

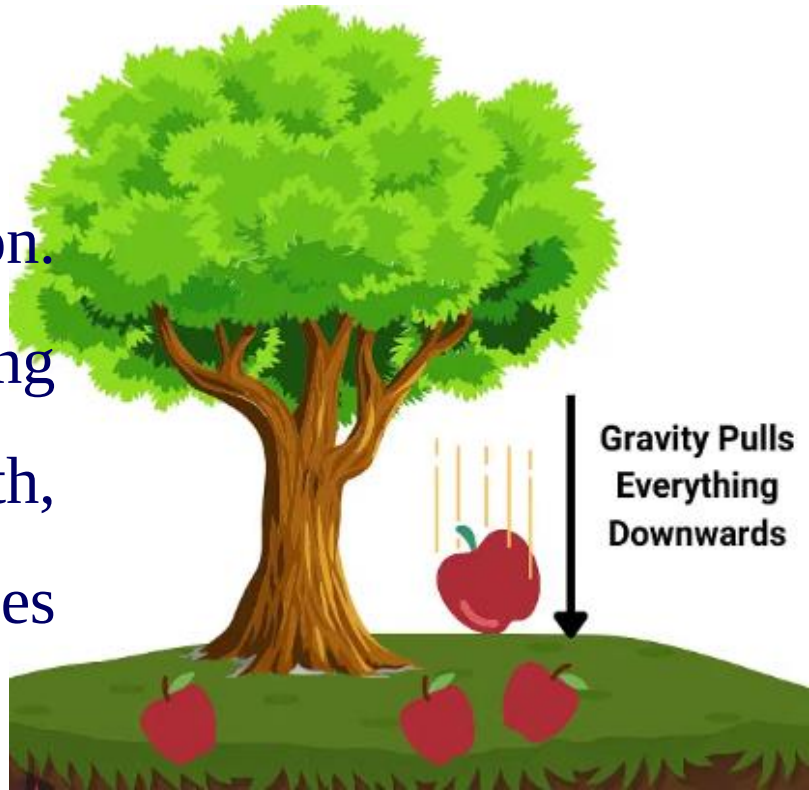
❖ The attractive force exerted by the Earth on an object is called the **gravitational force** F_g

❖ This force is directed toward the center of the Earth.

❖ It's magnitude is called the **weight** of the object.

$$F_g = mg$$

❖ weight varies with geographic location. Because **g** decreases with increasing distance from the center of the Earth, objects weight less at higher altitudes than at sea level.

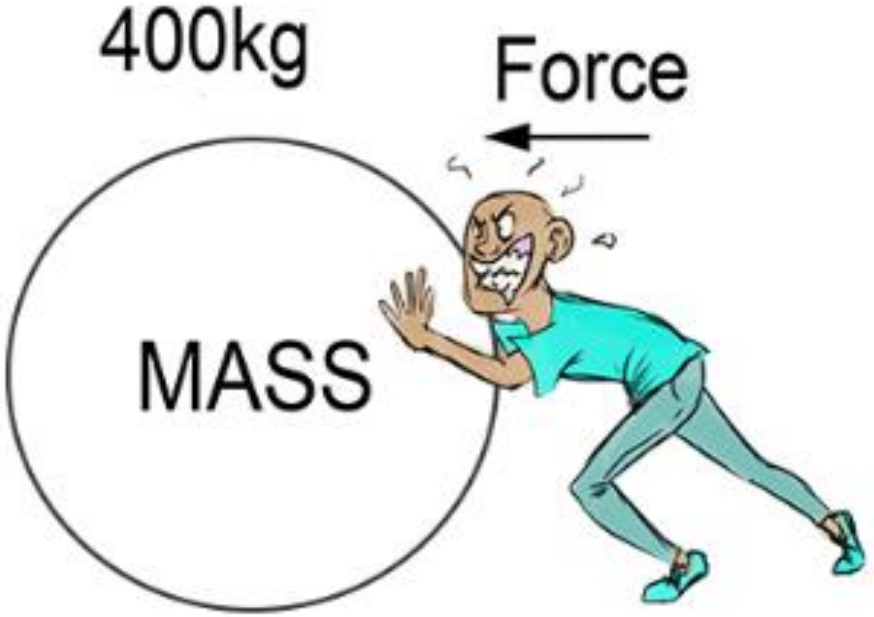


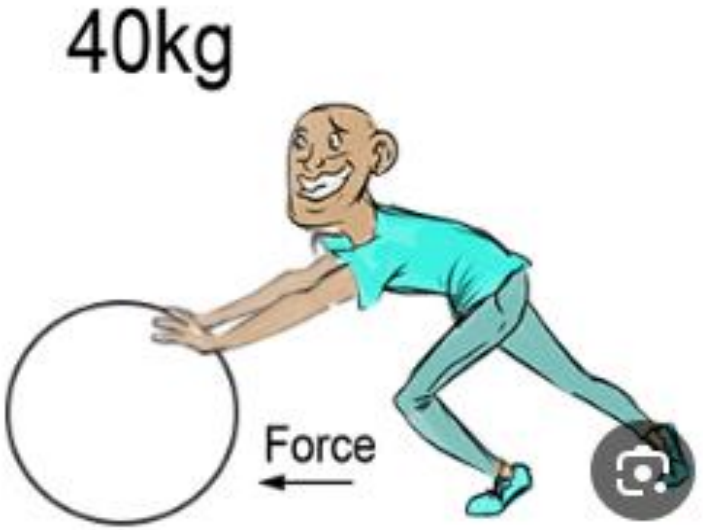
Newton's Second Law:

- It answers the question of what happens to an object that has a nonzero resultant force acting on it.

- The acceleration of an object is directly proportional to the force acting on it.

- The acceleration of an object also depends on its mass





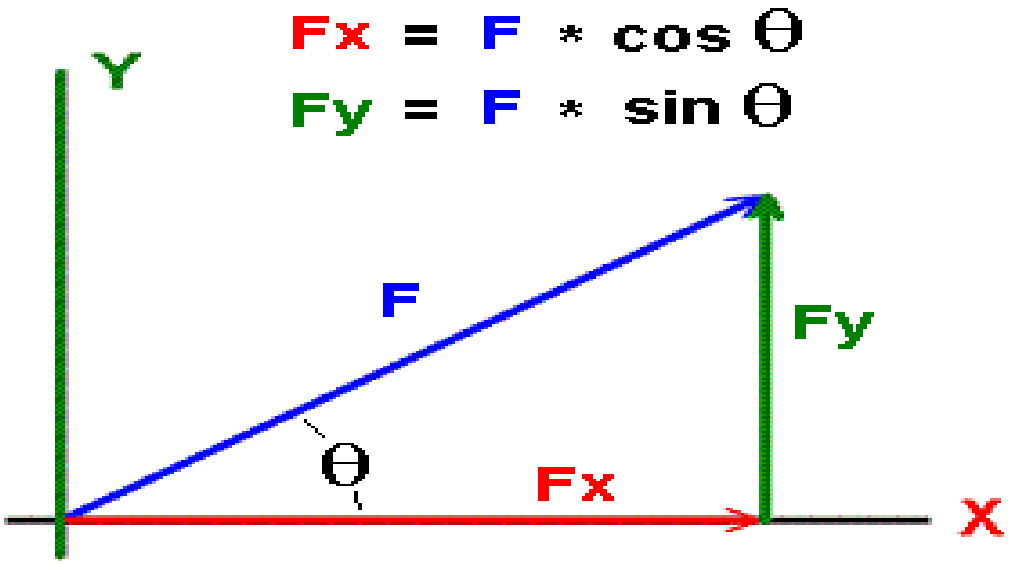
❖ Newton's Second Law states that :

" The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass"

$$\sum \mathbf{F} = m\mathbf{a}$$

❖ If there are many forces acting on an object, then

$$\sum F_x = ma_x \qquad \sum F_y = ma_y \qquad \sum F_z = ma_z$$



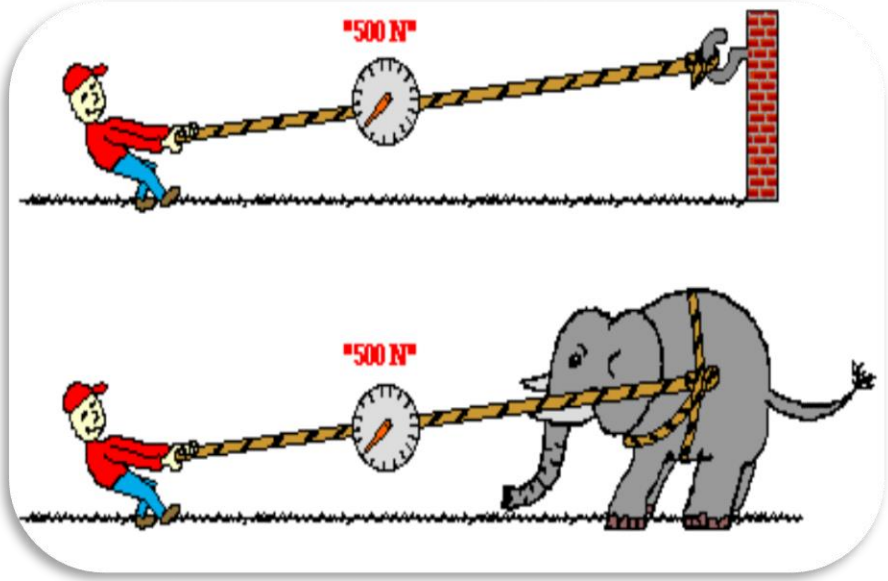
Newton's Third Law:

- It states that:

If two objects interact, the force F_{12} exerted by object 1 on object 2 is equal in magnitude and opposite in direction to the force F_{21} exerted by object 2 on object 1

$$\mathbf{F}_{12} = -\mathbf{F}_{21}$$

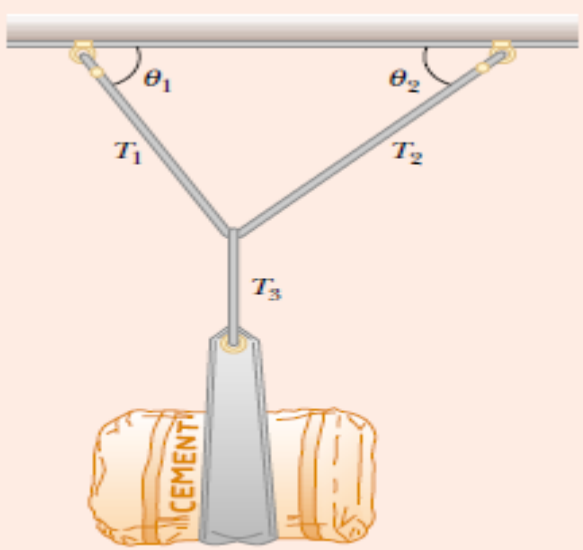
- The force that object 1 exerts on object 2 may be called **the action force** and the force of object 2 on object 1 **the reaction force**.





Example

A bag of cement of weight 325 N hangs from three wires as suggested in Figure P5.18. Two of the wires make angles $\theta_1 = 60.0°$ and $\theta_2 = 25.0°$ with the horizontal. If the system is in equilibrium, find the tensions T_1, T_2, and T_3 in the wires.



Solution

From equilibrium of the sack:

$$T_3 = F_g \qquad [1]$$

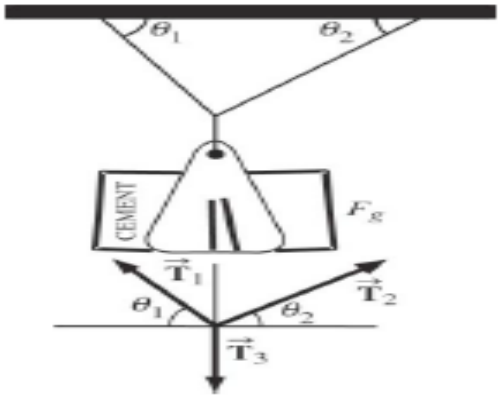
From $\sum F_y = 0$ for the knot:

$$T_1 \sin\theta_1 + T_2 \sin\theta_2 = F_g \qquad [2]$$

From $\sum F_x = 0$ for the knot:

$$T_1 \cos\theta_1 = T_2 \cos\theta_2 \qquad [3]$$

Eliminate T_2 by using $T_2 = T_1 \cos\theta_1 / \cos\theta_2$ and solve for T_1:



ANS. FIG. P5.33

$$T_1 = \frac{F_g \cos\theta_2}{(\sin\theta_1 \cos\theta_2 + \cos\theta_1 \sin\theta_2)} = \frac{F_g \cos\theta_2}{\sin(\theta_1 + \theta_2)}$$

$$T_3 = F_g = \boxed{325\ \text{N}}$$

$$T_1 = F_g\left(\frac{\cos 40.0°}{\sin 100.0°}\right) = \boxed{253\ \text{N}}$$

$$T_2 = T_1\left(\frac{\cos\theta_1}{\cos\theta_2}\right) = (253\ \text{N})\left(\frac{\cos 60.0°}{\cos 40.0°}\right) = \boxed{165\ \text{N}}$$


Thank
You